

Demand Forecasting System for Small Business Optimization

Mary Cherian, Aniket Jane, Deepak, G Sinchana, MD. Modasscer

Department of Information Science and Engineering

Dayananda Sagar Collage of Engineering,

Bangalore, India.

mary-ise@dayanandasagar.edu, aniketmjane@gmail.com, deepakosame888@gmail.com

sinchanabhandari9@gmail.com, mohammed.modasscer17@gmail.com

Abstract—Demand forecasting is a fundamental element in efficient inventory control and supply chain management through the ability of firms to anticipate future demand and devise methods to service that demand accordingly. The present paper explores two major approaches to demand forecasting: the SARIMA model and the Deep AR model. SARIMA is a more traditional statistical approach and exploits autoregressive, integrated, and moving average components along with seasonal adjustments to pick up trends and seasonality in the sales data. DeepAR is the most advanced deep learning-based probabilistic approach in modelling complex time series dynamics and relationships within interconnected time series. This paper details the process of demand forecasting starting from data collection, preparation, and cleaning stages to model building, testing, and validation. SARIMA was able to clearly prove its fit for short- to medium-term forecasting, enabling it to be very efficient in keeping track of inventory stocks and preventing stockouts. However, DeepAR does outperform SARIMA when using large dynamic datasets where non-linear patterns come into play, providing probabilistic forecasts that quantify uncertainty. Our findings have illuminated the strengths and weaknesses of both methods. Given stable, well-structured datasets, SARIMA is a robust choice. However, DeepAR's advanced capabilities are well-suited for businesses dealing with complex and dynamic conditions. Firms can optimize operations to adapt to market fluctuations and even make data-driven decisions to enhance efficiency and customer satisfaction by using these tools.

Keywords—Demand forecasting, SARIMA, DeepAR, time series analysis, inventory management, probabilistic forecasting.

I. INTRODUCTION

Demand forecasting is critical to modern business practice in order to maintain a competitive edge: it works together with efficient management of inventories and scheduling of production, and it controls supply chain activities. Being familiar with the meaning of demand forecasting is very crucial since it involves the intervals for which the demands for the particular goods or services are expected. Good forecasting prevents any costs associated with overstocking and understocking, improves customer service levels, and improves the efficiency of the supply chain. The demand forecasting enables businesses to make informed decisions that align resources with market needs.

II. FORECASTING MODELS AND MATHEMATICAL REPRESENTATION

Forecasting methods in time series analysis have been developed with the aim of improving and estimating the range

of predictions. Most of these methods are mainly used in the modelling of time-dependent processes thanks to the incorporation of structure: trends, seasonality, and other regimes present in the time series data.

A. SARIMA Model

1) SARIMA Model:

The Seasonal Auto Regressive Integrated Moving Average (SARIMA) model is a well-known time series forecasting method that handles both trend and seasonality in the data. It is mathematically expressed as:

$$SARIMA(p, d, q)(P, D, Q)s \quad (1)$$

Where:

- p, d, q (non-seasonal order):
- p : Number of lag observations used to predict the current value. Example: If $p = 2$, the model uses the past 2 values.
- d : Number of times the data is differenced to make it stationary. Example: If $d = 1$, the model uses the change between consecutive values.
- q : Number of past forecast errors used to predict the current value. Example: If $q = 1$, the model uses the previous error term.
- P, D, Q (seasonal order):
- s : Length of the seasonality cycle. Example: For monthly data with yearly seasonality $\rightarrow s = 12$, For weekly data with yearly seasonality $\rightarrow s = 52$
- P : Similar to p , but for the seasonal component, uses past seasonal values.
- D : Number of times the data is differenced at the seasonal level to remove seasonal trend. Example: If $D = 1$ and $s = 12$ (monthly data with yearly seasonality), the model uses the difference between this month and the same month last year.
- Q : Similar to q , but for seasonal errors.

SARIMA works very well for industries such as retail, manufacturing, and logistics that have a demand curve with a certain level of oscillation because of the prevailing seasonality, for short- to medium-range forecasting periods. However, these types of models often have a challenge in capturing the irregular swings and structural changes when there is an intricate structure contained in the database used even within dynamic markets.

2) DeepAR Model

$$Y_{t+1} = f(Y_t, Y_{t-1}, \dots, Y_1) \quad (2)$$

Where:

- Y_{t+1} is the forecasted value at the next step.
- Y_t, Y_{t-1}, Y_1 represent the sequence of past observed values, and the function f is a non-linear function learned by a recurrent neural network (RNN).

B. Combined Model:

$$y^t_{combined} = (y^t_{SARIMA} + y^t_{DeepAR}) / 2 \quad (3)$$

Where:

- $y^t_{combined}$: Combined forecast at time t ; the average of SARIMA and DeepAR predictions.
- y^t_{SARIMA} : Forecast at time t from the SARIMA model, which captures linear trends and seasonality.
- y^t_{DeepAR} : Forecast at time t from the DeepAR model, which learns non-linear and complex patterns. Simple averaging technique combines the outputs of both models equally.

The SARIMA model (1) works well for seasonal and linear patterns, while DeepAR (2) handles complex and changing data using deep learning. To get the best of both, we take the average of their predictions to create a combined forecast (3). This helps improve overall accuracy and makes the model more reliable.

The integration of traditional statistical methods such as SARIMA (1) with advanced deep learning models like DeepAR (2) leads to a hybrid forecasting framework that captures both seasonality and nonlinearity. The final ensemble forecast (3) benefits from the interpretability of SARIMA and the adaptability of DeepAR, offering improved accuracy and resilience for demand prediction in uncertain or highly variable environments.

III. LITERATURE SURVEY

[1] Tereza Šustrová et. al. from the Brno University of Technology, explored the challenges posed by the conventional inventory models. With this, they recommended the application of artificial neural networks (ANNs) due to the complexities posed by supply chains. The authors experimented with different configurations of ANNs, while the training function TRAINGDX and transfer function TANSIG were employed. This research applied data analysis and pre-processing with MS Office Excel and the model creation; testing was done with the Neural Network Toolbox of MATLAB software. In this study, the optimal model that had eight hidden neurons was very effective ($R^2 = 0.99$) and had very low Mean Squared Error ($MSE = 0.029$) and found that Managing Neural Networks can help even to determine how much order to place in order to enhance management of order cycle dealing with inventory levels problems.

[2] Maryam Zohdi et al. of Sharif University of Technology and Hamad bin of Khalifa University have put forward the idea of advancing demand prediction by machine learning algorithms. The authors discussed the limitations of oft-used statistical approaches such as ARIMA when applied

to complex non-linear data. Following this, they, perhaps excitedly, started looking up some of the more advanced approaches like Extreme Learning Machine (ELM), Decision Trees, Gradient Boosting, Multi-Layer Perceptron (MLP), K-nearest Neighbors (KNN), and so on. The models used acquired data from a marketing campaign referred to as Black Friday, with some customer inclination and situation related assessment criteria, Mean Squared Error (MSE) and Mean Adjusted Error (MAE), etc. The outcomes showed that ELM and MLP were in all likelihood better than traditional approaches meaning that they could be depended upon as far as maps of their structure were concerned.

[3] José Eduardo de Carvalho Lima et al. made a systematic review that provides some areas for improvement regarding demand forecasting and planning of production systems, but it is not very useful practically since it does provide a comparative assessment of different models. On the background of the study, some models are very complex and therefore, applicability can be a challenge. In this regard, the scope of research is imbalanced as some crucial areas remain inadequately addressed. In any case, the paper is helpful in identifying weaknesses of the existing research, but it could have improved its contributions through more comprehensive treatment of some particular techniques used.

[4] Aditya Chawla et al. analysed applications of ANNs in demand forecasting. Scenarios presented the capability of ANNs in demand forecasting. Results thereby attained reflected satisfying for using ANNs in forecasting, mainly in the integration of complex data patterns. The huge computational costs involved with models developed using ANNs will be one major drawback to this kind of model. The computational complexity does cause a particular problem when very large data sets are involved, since tools like MATLAB, unfortunately, tend to limit the ability to scale up models that would accommodate large volumes of data satisfactorily.

[5] Muhammad Adnan Khan et al. at Lahore Garrison University designed and implemented a business intelligence driven demand forecasting system which integrates machine learning into its workings. Once they applied the deepAR algorithm on AWS Sage maker, they could analyse sales and inventory data with up to 92.38% accuracy. Most importantly, this model can be of substantial assistance to companies those want to reduce wasteful production and improve the utilization of available resources, especially those in need of new products but have little data to consult.

[6] Real Carbonneau et al., of Concordia University, studied some machine learning techniques, particularly NN, RNN, and SVM, in addressing bullwhip effect in supply chain demand forecasting. These advanced techniques were then compared with traditional methods in the form of moving averages and regression models with both simulated data and real data. Though RNNs and SVMs have shown an improvement in the accuracy of their forecasting, the differences weren't statistically significant. SVMs provided a solid mathematical base to excel in dealing with complex time series with minimum overfitting. Therefore, the use of machine learning is expected to increase the quality of forecasts in order to streamline the supply chain.

[7] Fattah et al. investigated demand forecasting using the ARIMA model in a Moroccan food company, employing the Box-Jenkins methodology with historical data from 2010–

2015. They developed and evaluated several ARIMA models based on criteria like AIC, SBC, and maximum likelihood, selecting ARIMA (1, 0, 1) as the most suitable. The chosen model demonstrated high accuracy in predicting demand, facilitating improved production planning, inventory management, and cost reduction. This study highlighted ARIMA's effectiveness in forecasting demand for perishable goods, ensuring better decision-making and enhanced supply chain efficiency.

[8] Naoufel Cheikhrouhou et al. performed a review on collaborative forecasting that combines mathematical models with event-based fuzzy judgments. The study highlights the problems of using these methods because of different perceptions by the forecasters and the difficulty of integrating contextual knowledge. Although it emphasizes collaboration within firms, it pays less attention to inter-firm processes, thus creating a gap in the overall analysis. The research explains the limitations of conventional forecasting models and proposes systematized solutions to improve predictions. However, it does not offer a critical comparison of various techniques available and, therefore is less useful practically. Nevertheless, the study outlines important insights related to the combining of human judgement and models with demand forecasting within changing environments.

[9] Fatima Zohra Benhamida et al. published a review paper on demand forecasting methods for time-series data. The paper was proposing a novel demand forecasting tool for inventory control. In such a study, the strength in integrating statistical and machine learning methods is presented. However, such a study is lacking in practical comparative insights into its application across diversified industries. Given the background of the study, the complexity in hybrid methods gives rise to problems in adoption. Additionally, the scope is somewhat imbalanced because it focuses less on the area of intermittent demand forecasting and new product introduction forecasting. The paper identifies substantial gaps and is very helpful for developing robust tools for forecasting.

[10] Kanik et al. investigated the application of regression-based machine learning techniques for forecasting sales within supply chain operations. The study focused on leveraging predictive analytics to analyse sales data, identifying demand trends and patterns. Their findings highlighted how regression models could support supply chain optimization and enhance inventory management decisions.

[11] Sukolkit et al. proposed an innovative approach to inventory management that integrates demand forecasting techniques specifically tailored for the steel industry. The study emphasized the importance of open innovation in addressing inventory challenges, utilizing advanced forecasting methods to predict demand trends accurately. Their research provided insights into optimizing inventory management processes in a dynamic and complex industrial environment.

[12] Khan et al. (2020) introduced a demand forecasting model enhanced by business intelligence and machine learning techniques. The study highlighted the integration of machine learning algorithms to improve the accuracy and efficiency of demand predictions. By leveraging business intelligence tools, the model provided actionable insights, supporting strategic decision-making processes across various business operations.

IV. PROPOSED SYSTEM

A. Data Source

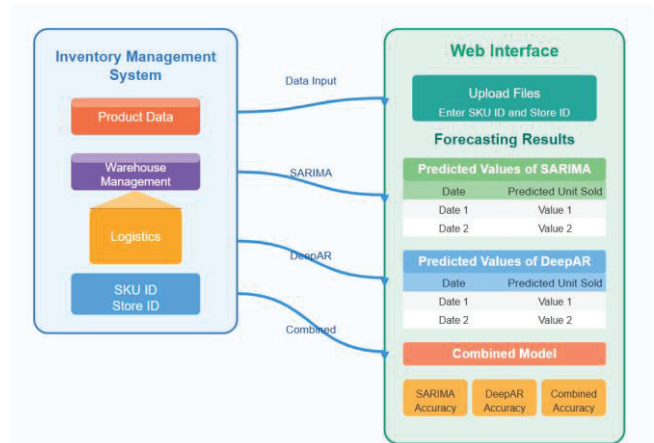


Fig. 1. Overview of System Data Source

Figure 1 depicts the framework that links an inventory management system to a web interface. Product and logistics data are processed using SARIMA, DeepAR and the combined model to generate individual forecasts.

The proposed system relies on two key data sources: sales data and inventory data, both of which play a vital role in generating accurate predictions for unit sales. Each dataset is categorized using unique Stock Keeping Unit identities (SKU IDs) and Store identities, ensuring that the forecasts are highly specific to individual products and their respective store locations. Historical sales data captures trends in customer demand, while inventory data ensures that predictions are aligned with stock availability, helping to avoid overstocking or stock shortages and help in the process of inventory management.

This data is uploaded into a web interface where it undergoes preprocessing, such as cleaning and formatting, to prepare it for machine learning analysis. Advanced algorithms are then applied to identify patterns and correlations between various factors influencing demand. By combining real-time and historical data labelled with SKU IDs and Store IDs, the system produces reliable, product-specific, and location-based forecasts. The system can also incorporate external factors, such as seasonal trends or market shifts, to enhance prediction accuracy further. This comprehensive approach enables businesses to make informed decisions, streamline inventory management, and improve overall operational efficiency, ultimately boosting customer satisfaction.

B. Machine Learning Models

1) SARIMA AND DEEPAR:

The SARIMA model is a statistical method of time series forecasting aimed at data modelling that follows the seasonal patterns. Its methodology is essentially an extension of the ARIMA model, taking the seasonal into account, and is particularly well-suited for datasets in which seasonality makes a significant difference to demand. The three components on which SARIMA is based are Autoregressive, Integrated, and Moving Average, with their seasonal counterparts. Notation SARIMA (p, d, q) (P, D, Q, m) where p, d, and q depict the non-seasonal AR, differencing and MA terms. Their seasonal equivalents, P, D, Q, and m with m as

the number of observations in each season are denoted, The SARIMA method focuses on an exhaustive analysis of the data to work out the underlying trends, seasonality, and noise. This involves exploratory data analysis, where visualizations and various statistical tests, such as the autocorrelation and partial autocorrelation functions, help identify possible seasonal lags and check for stationarity. Stationarity is a crucial assumption in SARIMA modelling, since it ensures that the statistical properties of the time series, mean and variance, are invariant with respect to time. In the case of non-stationary in data, differencing is done. Seasonal differencing is specially used to remove periodic fluctuations, while regular differencing removes overall trends.

This paper will focus on the usage of a robust methodology for forecasting demand through SARIMA and DeepAR models and a stronger methodology for accurate and more interpretable results. The dataset has historical sales data supplemented with economic indicators and seasonal trends to capture demand fluctuations comprehensively. Data was sourced from internal company repositories and external publicly available databases. To ensure sufficient coverage of patterns and anomalies, a daily granularity spanning five years is chosen, offering both short-term and long-term perspectives critical for effective forecasting.

Before model application, rigorous data pre-processing steps are undertaken. Missing values, a common challenge in time series datasets, are handled using linear interpolation, preserving the temporal continuity essential for accurate forecasting. Outliers could skew model predictions. They are detected using interquartile ranges and replaced with median values, hence upholding the integrity of the dataset. The variance of the data is stabilized through log scaling, and differencing is used to ensure stationarity; this is critical for models as complicated as SARIMA. These pre-processing steps ensure that the dataset is optimized for reliable analysis and modelling.

For model choice, both SARIMA and DeepAR are chosen for their complementary strengths in handling seasonal and non-linear time series data. SARIMA is most suited for

datasets exhibiting clear seasonal patterns, as it can include autoregressive, moving average, and seasonal components. This makes it strong in modelling periodic demand variations. However, DeepAR, a probabilistic forecasting model that uses recurrent neural networks (RNN), proves superior in grasping complex dependencies present in the data. The ability of deep learning models to deal with covariates, and hence outputting probabilities for the underlying predictions, is extremely helpful in dynamic patterning of missing data.

As shown in figure 2, it depicts a forecasting pipeline with data preprocessing, feature engineering, and two model SARIMA (statistical) and DeepAR (deep learning). Their outputs are fused using an ensemble approach to generate the final forecast.

The train is split into 70% for training, 20% for validation, and 10% for testing so as to allow robust model evaluation and generalization. SARIMA's parameters, comprising autoregressive (p), differencing (d), and moving average (q) components, together with seasonal counterparts (P, D, Q, m) are selected with grid search and the model is assessed with Akaike Information Criterion (AIC). In DeepAR, hyper parameters like the number of RNN layers, hidden units, and learning rate are tuned with Bayesian search to ensure the model is both efficient and accurate. Model performance is evaluated using several metrics, namely Mean Absolute Error, Mean Squared Error, and Mean Absolute Percentage Error. Mean Absolute Error gives a simple average measure of deviation, while the Mean Squared Error punishes larger errors, making it more sensitive to significant deviations in forecasts.

Mean absolute percentage error is particularly useful in providing an intuitive interpretation of the accuracy of forecasts relative to observations. These metrics collectively ensure a balanced assessment of model efficacy across different dimensions of error sensitivity and interpretability. The final choice of model is guided by comparisons on the validation dataset. Both SARIMA and DeepAR are compared to the evaluation metrics defined. SARIMA's interpretability together with its statistical basis is compared with the ability of DeepAR to model non-linear dependencies as well as the capability of integrating covariates. In situations where both have similar performance, operational factors, such as computational efficiency and scalability, play a decisive role. The model with the minimum average Mean absolute percentage error over validation datasets is preferred for deployment with consideration of practical implementation aspects, aligning with business objectives. With this methodology, then, the SARIMA and DeepAR are combined in order to provide robust, accurate, and interpretive demand forecasts. Thus, by incorporating rigorous data pre-processing, systematic parameter tuning, and comprehensive evaluation, this approach will ensure that the model chosen is not only technically sound but also operationally viable. It will allow businesses to make informed decisions and optimize inventory levels and, hence, improve general supply chain efficiency.

Moreover, businesses must ensure that their forecasting models are adaptable to market changes and external disruptions. Regular updates to the models using the latest data can significantly improve their accuracy. Additionally, integrating domain knowledge with machine learning techniques can enhance model interpretability and decision-making. Future research can explore hybrid approaches

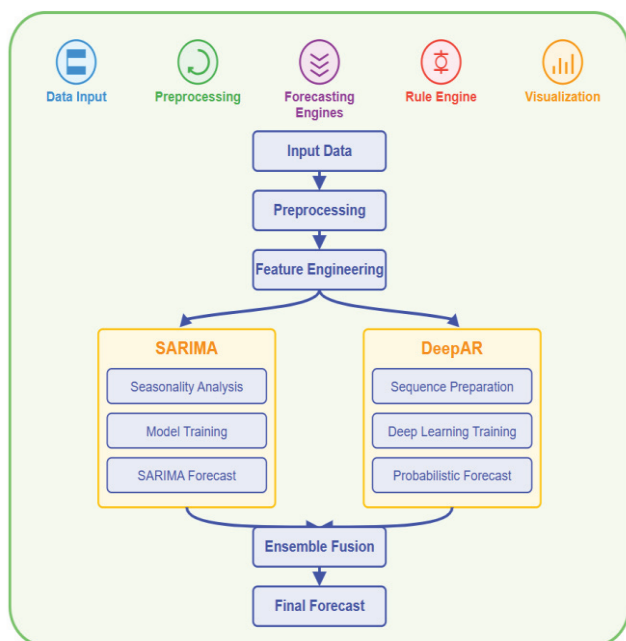


Fig. 2. System Architecture

incorporating real-time external factors such as economic indicators and competitor strategies. This would further enhance forecasting accuracy and operational efficiency, ultimately leading to better business outcomes.

V. SIMULATION RESULTS

This simulation demonstrates demand forecasting using two models, SARIMA and DeepAR, to predict product sales for a specific SKU (216419) at a particular store (Store ID: 8091). The interface allows users to upload data, set a prediction horizon (default of 12 weeks), and input store and SKU details. The SARIMA model, a traditional statistical approach, provides predictions with an accuracy of 50.76%, reflecting its limitation in capturing complex temporal patterns and irregularities. In contrast, the DeepAR model, which is based on deep learning, achieves a much higher accuracy of 81.0%, indicating its ability to model time-series data and identify trends a superior manner.

To further enhance reliability, a combined model is created by merging predictions from both SARIMA and DeepAR, resulting in an overall accuracy of 77.14%. This is a hybrid approach that balances the strengths of both models, thereby offering a robust forecasting solution. All three models predicted values are presented with accuracy metrics to allow comparisons. A graph visualizes the combined forecast over the prediction period, illustrating sales trends and fluctuations for better decision-making. By integrating visualization tools, the interface provides actionable insights, enabling businesses to respond proactively to demand fluctuations. The ability to fine-tune parameters for both models ensure flexibility and adaptability to diverse forecasting scenarios, enhancing decision-making and operational efficiency. This simulation, therefore shows how effective advanced deep learning approaches, such as DeepAR, are in demand forecasting and the importance of amalgamation of traditional and modern approach towards achieving improved accuracy metrics to allow comparison.

Upload Files

Select File here

Files Supported: CSV

Choose File

No of Weeks to predict (default is 12)

Enter Store Id

Enter Sku Id

Predict

Fig. 3. Web Interface

As shown in figure 3, the interface allows users to upload a CSV file, specify the number of weeks for prediction, and enter store and SKU IDs before generating a forecast.

SKU id:216419 Store id:8091											
Predicted Values of SARIMA				Predicted Values of DeepAR				Predicted Values of Combined Model			
Date	Predicted Unit Sold			Date	Predicted Unit Sold			Date	Predicted Unit Sold		
2013-09-04	18.0			2013-09-04	17.0			2013-09-04	18.0		
2013-09-07	20.0			2013-09-07	13.0			2013-09-07	16.0		
2013-11-06	24.0			2013-11-06	9.0			2013-11-06	16.0		
2013-12-02	16.0			2013-12-02	20.0			2013-12-02	18.0		
2013-12-03	8.0			2013-12-03	19.0			2013-12-03	13.0		
SARIMA Accuracy				DeepAR Accuracy				Combined Accuracy			
50.76 %				81.0 %				77.14 %			

Fig. 4. Predicted Sales Comparison Across Models

The table in figure 4, compares predicted unit sales from SARIMA, DeepAR, and their combined model for a specific SKU and store. Accuracy improves from SARIMA (50.76%) to DeepAR (81.0%), with the combined model achieving 77.14% accuracy.

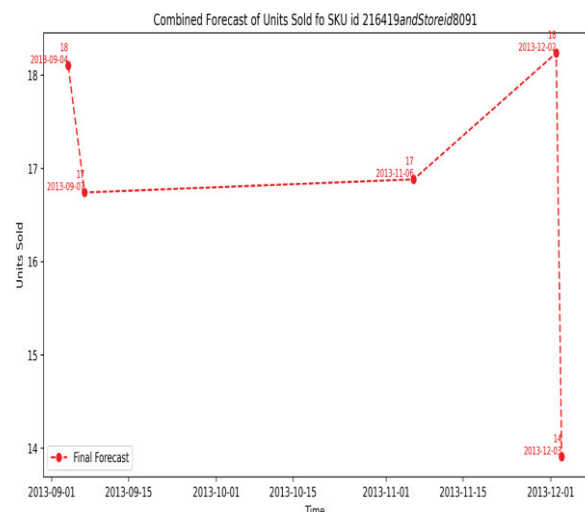


Fig. 5. Prediction Graph Interface

As shown in figure 5, the graph visualizes the final forecasted unit sales over time for a specific SKU and store, based on an ensemble of SARIMA and DeepAR model.

The ability to fine-tune parameters for both models ensures flexibility and adaptability to diverse forecasting scenarios, enhancing decision-making and operational efficiency. This simulation, therefore shows how effective advanced deep learning approaches, such as DeepAR, are in demand forecasting and the importance of amalgamation of traditional and modern approach towards achieving improved accuracy metrics to allow comparison. Future research can explore hybrid approaches incorporating real-time external factors such as economic indicators and competitor strategies. This would further enhance forecasting accuracy and operational efficiency, ultimately leading to better business outcomes.

VI. CONCLUSION

This research explores the approach of creating a demand forecasting system using the strengths of two powerful models: SARIMA and DeepAR, for predicting future sales in the retail environment. The study analysed SARIMA and DeepAR models for demand forecasting in small businesses. SARIMA performed well for structured, seasonal data but struggled with complex patterns, whereas DeepAR, a deep learning approach,

achieved higher accuracy (81.0% vs. SARIMA's 50.76%). A hybrid model combining both improved accuracy to 77.14%, demonstrating the benefits of integrating statistical and machine learning techniques.

The demonstration showed how such systems can deal with seasonal trends and even empirical multivariate data, resulting in accurate and reliable forecasts of retail demand. The evaluation of the hybrid forecasting model indicates improved prediction accuracy over standalone models for small business inventory forecasting. The other key advantages of the system are scalability and efficiency. It could process quite large volumes of highly needed data within reasonable time limits, thus being ideal for high-volume retail environments. The user-friendly system allows singular stakeholders to load data and understand what was forecasted using clarity in visualizations and accuracy metrics. The system thus stands as an efficient tool for inventory and sales planning where retailers can optimize stock levels. This data-driven forecasting tool can help small businesses minimize waste, streamline operations, and make more informed decisions for sustainable growth.

VII. FUTURE WORK

While the proposed demand forecasting system performs effectively, there are several avenues for improvement and expansion. One of the primary areas for enhancement is the incorporation of external variables, such as promotions, holidays, weather conditions, and economic indicators. These factors can significantly influence demand patterns, and integrating them into the forecasting models could further improve the accuracy and robustness of predictions. This will result in more precise and context-aware forecasts, making it better suited for dynamic and unpredictable retail environments. Another direction for future work is to refine the model to handle larger datasets more efficiently. Additionally, integrating real-time data processing would allow continuous updating of predictions based on current sales trends, improving short-term forecasting accuracy and enabling quicker decision-making for retailers. Combining SARIMA and DeepAR with XGBoost or LSTM (Long Short-Term Memory), could improve the system's ability to handle irregular sales patterns or highly volatile demand. Furthermore, providing users with the ability to customize model settings and select relevant external variables would increase the flexibility of the system, making it adaptable to a wider range of retail scenarios and business needs.

Future work can also explore external factors like economic trends. Advanced forecasting techniques will

empower small businesses with data-driven insights for efficiency and growth.

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